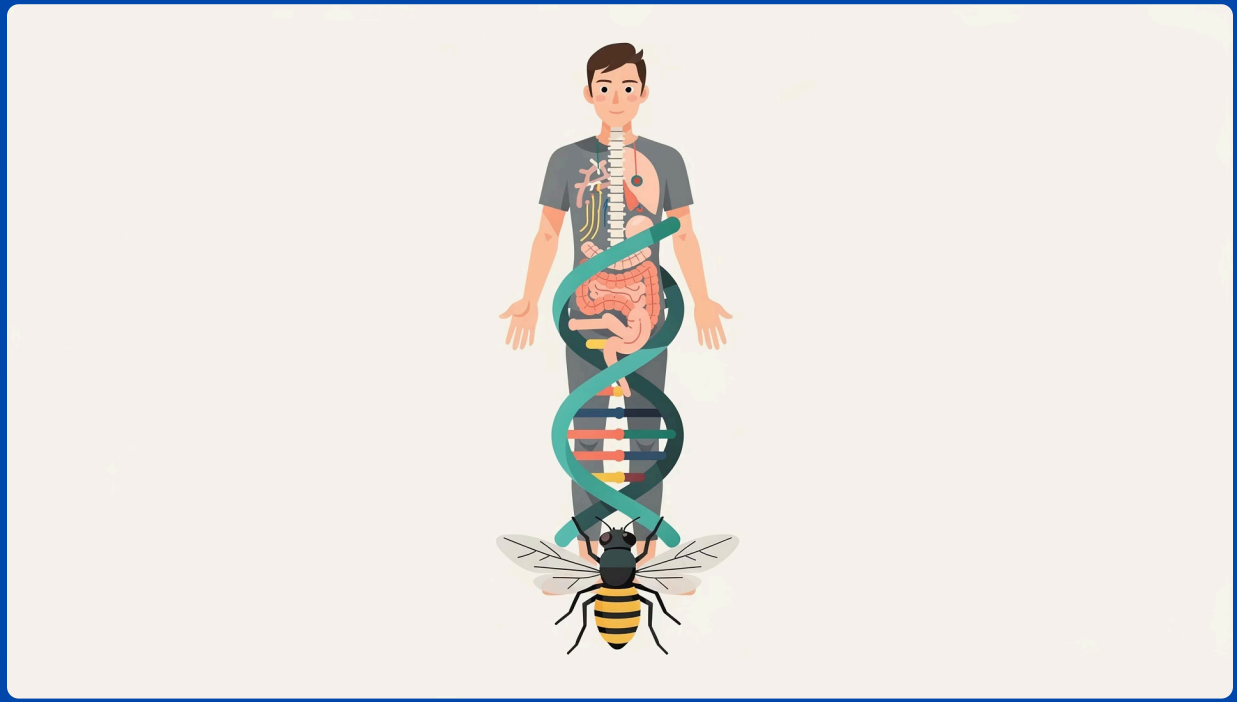
Who Gives a Fly

Bridging the valley of death between model organism genetics and human clinical outcomes

CONTEXT

Why This Matters



The Translation Gap

Genetic discoveries in flies often fail to translate to human therapies

Clinical Failure Rate

90% of drugs fail between model testing and human trials

Cross-Species Learning

Understanding conserved pathways accelerates drug development

CASE STUDY

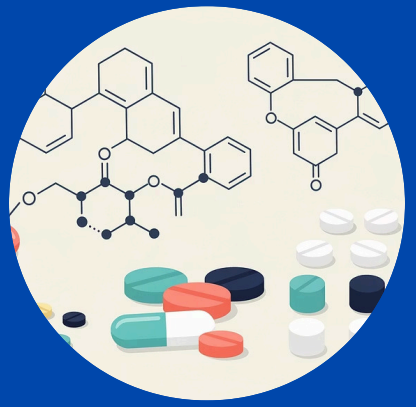
The Rb Pathway as Universal Language

Geneticists

Ancient cell cycle control mechanisms conserved across species

Oncologists

Retinoblastoma protein dysfunction drives tumor formation

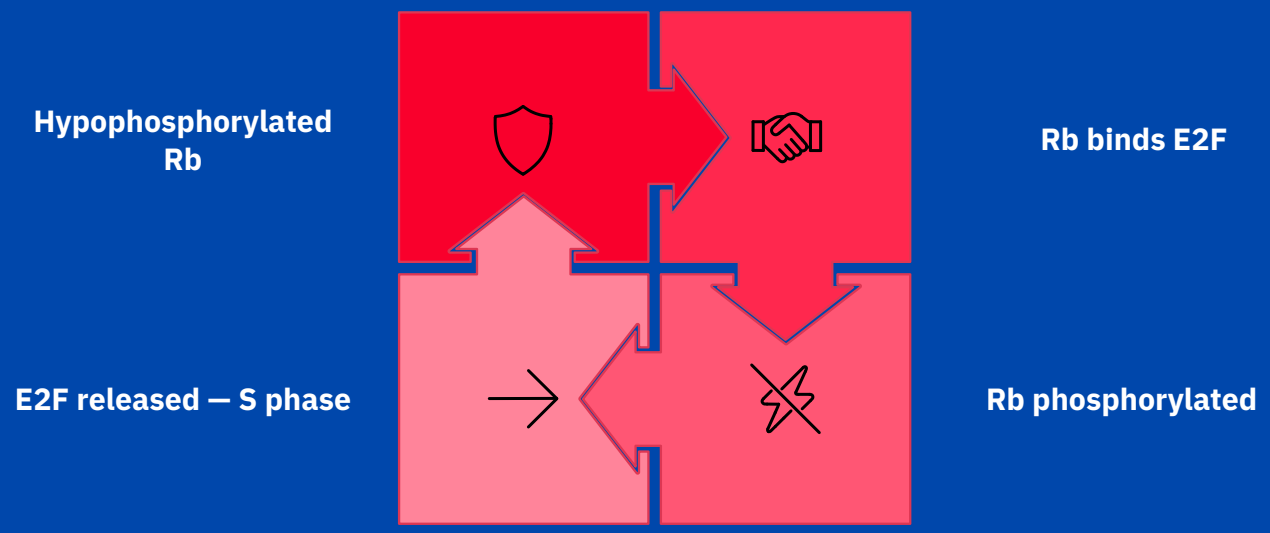


Pharmacologists

Target validation across evolutionary conserved pathways

Core cell cycle machinery provides a common framework for interdisciplinary collaboration

The Rb Pathway: Cross-Species Conservation



The Molecular Brake

Active state: Hypophosphorylated Rb binds E2F transcription factors

Result: Cell cycle arrest at G_1/S checkpoint

Conserved across flies, mice, and humans

Identifying the "Gaps" for Your Project

Three critical categories where model organisms diverge from human biology

GAP #1

Redundancy and Paralogs

The Challenge

Humans possess backup systems that flies lack

Humans & Mice

Rb family triad: $RB1$, $RBL1/p107$, $RBL2/p130$

Drosophila

Single RBF gene – less redundancy, clearer phenotypes

A drug showing promise in flies may be "buffered" by p107/p130 in humans, causing clinical trial failure

GAP #2

Tissue Specificity

Species-Specific Vulnerability

- **Humans:** *RB1* mutations → retinal tumors during development
- **Flies:** *RBF* loss → generalized overgrowth (wing, eye discs)
- **Mice:** Intermediate phenotypes, different tissue sensitivities

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Why do certain tissues respond differently? This question is critical for cross-disciplinary drug development teams.

GAP #3

Metabolic Scaling

> Genetic homology doesn't guarantee pharmacological equivalence

A molecule stabilizing Rb in flies may have drastically different:

- Half-life (minutes vs. hours)
- Tissue distribution
- Hepatic metabolism
- Bioavailability

SUMMARY

Key Takeaways

Conservation enables discovery

Rb pathway homology provides powerful starting point for genetic insights

Gaps predict failure

Redundancy, tissue specificity, and metabolic scaling explain translation barriers

Education bridges disciplines

Cross-training in these gaps accelerates therapeutic development

Next Steps

Educational Materials Development

01

Video Series

Animated explainers on pathway conservation and divergence

02

Interactive Modules

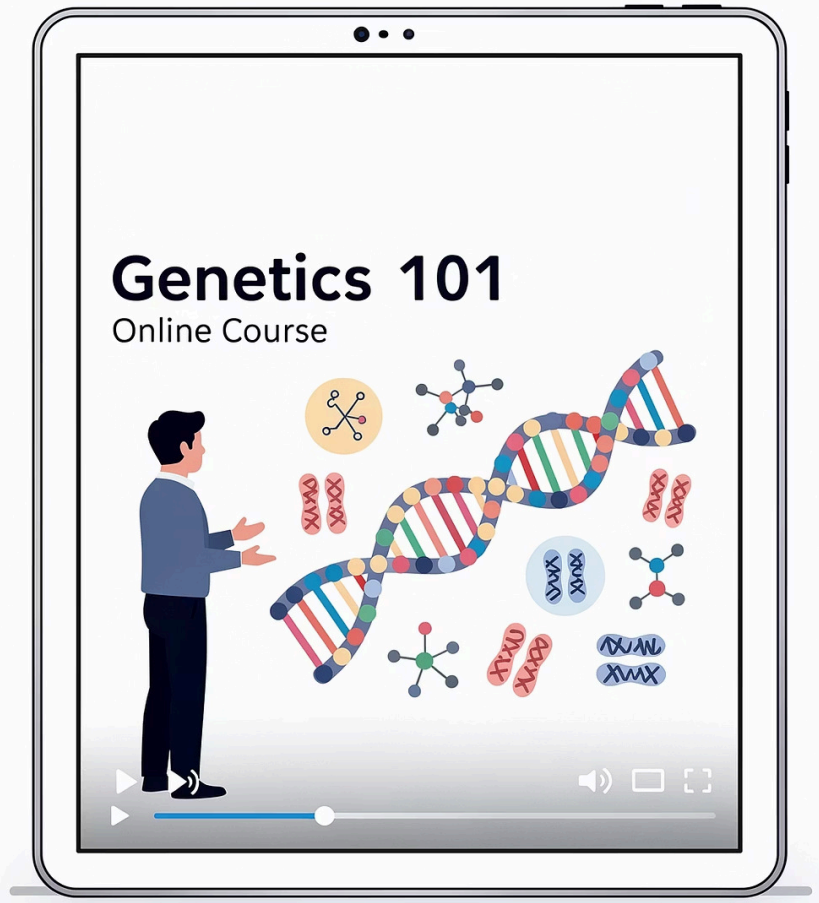
Case studies comparing fly, mouse, and human Rb biology

03

Cross-Disciplinary Workshops

Collaborative training for research teams

Goal: Equip biomedical teams with tools to anticipate and address translation gaps early in the drug development pipeline



Conclusions

Model organisms remain invaluable

Flies provide rapid, cost-effective genetic insights impossible in mammals

Translation requires expertise

Understanding molecular, cellular, and pharmacokinetic gaps is essential

Education drives innovation

Cross-disciplinary literacy transforms how we bridge the valley of death

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Who gives a fly? Anyone serious about translating genetic discoveries into human therapies.